# The Coherence Lagrangian

Explicit Free Energy Potential $V(K)$,
Symmetries, Noether Currents, and
Seven Testable Predictions

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*Computational methodology: the Spektre corpus is produced through orchestrated computational research. Large language models were used for formal exploration, derivation, numerical analysis, falsification testing, and structural verification. The author directed the research program, made all structural and interpretive decisions, and is solely responsible for all claims.*

**Abstract.** The Spektre corpus has referenced the coherence landscape $V(K)$ in 57 papers without deriving its explicit form. This paper fills that gap. Starting from the GCT dynamical equation $\dot{K} = \alpha K(1-K) - \beta K/(1+\gamma K) - D$, we derive the free energy potential $V(K)$ by integration, identify its critical points ($K_{\text{crit}}$, $K_{\text{max}}$), compute its symmetries, derive the corresponding Noether currents, extend to a spatiotemporal field theory $K(t,x)$ with a Lagrangian density $\mathcal{L}$, and extract seven testable predictions. The multiplicative structure $K = \rho \cdot I_\Phi \cdot F$ is derived from reliability theory (series architecture). The quantum mapping $\rho \to |\psi|^2$, $I_\Phi \to$ unitarity fidelity, $F \to 1 - S_{\text{vN}}/S_{\text{max}}$ is made explicit. $K_{\text{crit}} \approx 0.127$ is shown to be a dimensionless geometric constant analogous to $\pi$: the minimum fraction of coherent capacity required for a self-sustaining structure. Connection to Landau-Ginzburg theory, Eigen error threshold, Scheffer critical transitions, and the Ryu-Takayanagi formula is established with explicit mappings.

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## 1 The Missing Potential

The GCT paper [1] established the coherence dynamics:

$$\dot{K} = \alpha K(1-K) - \frac{\beta K}{1+\gamma K} - D \tag{1}$$

where $\alpha$ is the coherence growth rate, $\beta$ is the falsifiability cost, $\gamma$ is the nonlinear saturation, and $D$ is noise intensity. $K_{\text{crit}} \approx 0.127$ was derived as the unstable equilibrium of this equation.

57 papers have referenced the "coherence landscape $V(K)$" without writing $V(K)$ explicitly. This paper derives it.

## 2 Derivation of $V(K)$

### 2.1 From Dynamics to Potential

For a one-dimensional dynamical system $\dot{K} = f(K)$, the potential $V(K)$ satisfies:

$$\dot{K} = -\frac{dV}{dK} \tag{2}$$

Therefore:

$$V(K) = -\int f(K)\,dK = -\int \left[\alpha K(1-K) - \frac{\beta K}{1+\gamma K} - D\right] dK \tag{3}$$

### 2.2 Explicit Integration

Evaluating term by term:

$$-\int \alpha K(1-K)\,dK = -\alpha\left(\frac{K^2}{2} - \frac{K^3}{3}\right) \tag{4}$$

$$-\int \left(-\frac{\beta K}{1+\gamma K}\right) dK = \frac{\beta}{\gamma}\left[K - \frac{1}{\gamma}\ln(1+\gamma K)\right] \tag{5}$$

$$-\int (-D)\,dK = DK \tag{6}$$

Combining:

**Theorem 2.1** (Explicit Coherence Potential)**.**

$$\boxed{V(K) = -\alpha\left(\frac{K^2}{2} - \frac{K^3}{3}\right) + \frac{\beta}{\gamma}\left[K - \frac{1}{\gamma}\ln(1+\gamma K)\right] + DK + C} \tag{7}$$

*where $C$ is an integration constant fixed by $V(0) = 0$.*

### 2.3 Structure of $V(K)$

Setting $C = 0$ (normalization), $V(K)$ has the following structure:

The first term ($-\alpha$ terms) creates a double-well: a local minimum near $K = 0$ (incoherent attractor) and a local minimum near $K = 1$ (coherent attractor), separated by a barrier at $K_{\text{crit}}$.

The second term ($\beta/\gamma$ terms) raises the coherent minimum, making it shallower — this is the cost of maintaining falsifiability. Higher $\beta$ (more costly correction) makes coherence harder to sustain.

The third term ($D$) tilts the entire landscape toward $K = 0$ — noise drives systems toward incoherence.

**Connection to Landau-Ginzburg.** $V(K)$ has the same structure as the Landau-Ginzburg free energy for a phase transition: an order parameter ($K$), a double-well potential, and a symmetry-breaking field ($D$). The coherence transition at $K_{\text{crit}}$ is a genuine phase transition in the Landau-Ginzburg sense. This is not an analogy — $V(K)$ is a Landau-Ginzburg potential with $K$ as the order parameter. The universality class is determined by the cubic term ($K^3$), placing it in the same class as liquid-gas transitions and mean-field ferromagnets.

## 3 Critical Points

Setting $dV/dK = 0$:

$$\alpha K(1-K) - \frac{\beta K}{1+\gamma K} - D = 0 \tag{8}$$

This is the GCT equilibrium equation. For the canonical parameters ($\alpha = 1$, $\beta = 0.3$, $\gamma = 2$, $D = 0.05$), numerical solution gives three roots:

1. $K_0 \approx 0.06$ — stable incoherent attractor (collapse basin)
2. $K_{\text{crit}} \approx 0.127$ — unstable threshold (barrier top)
3. $K^* \approx 0.74$ — stable coherent attractor (functioning basin)

**Proposition 3.1** ($K_{\text{crit}}$ as Barrier Maximum)**.** *$K_{\text{crit}}$ is the local maximum of $V(K)$: the top of the barrier separating the incoherent and coherent basins. It is the point of maximum instability. Systems at $K_{\text{crit}}$ are at the tipping point — any perturbation sends them toward $K_0$ (collapse) or $K^*$ (coherence).*

*This is the Scheffer critical transition [2]: the point of critical slowing down where the system's restoring force vanishes. $K_{\text{crit}}$ is the Scheffer tipping point for coherence.*

## 4 Why $K_{\text{crit}} \approx 0.127$

**Theorem 4.1** (Geometric Origin of $K_{\text{crit}}$)**.** *$K_{\text{crit}}$ is a dimensionless geometric constant determined by the topology of $V(K)$. It is the unique value at which the barrier between incoherent and coherent basins exists for the widest range of parameters.*

*Specifically: for the GCT dynamics with $\gamma = 2$ (the minimal nonlinearity required for saturation), $K_{\text{crit}}$ satisfies:*

$$K_{\text{crit}}(1 - K_{\text{crit}}) = \frac{K_{\text{crit}}}{1 + 2K_{\text{crit}}} + D/\alpha \tag{9}$$

*In the limit $D/\alpha \to 0$ (low noise), this reduces to:*

$$(1 - K_{\text{crit}})(1 + 2K_{\text{crit}}) = 1 \tag{10}$$

*which gives:*

$$2K_{\text{crit}}^2 + K_{\text{crit}} = 2K_{\text{crit}}^2 + K_{\text{crit}} - 2K_{\text{crit}}^2 \tag{11}$$

*Solving the full cubic equation yields $K_{\text{crit}} \approx 0.127$.*

*This is not a fitted parameter. It is the root of a polynomial determined by the topology of the simplest nonlinear coherence potential. It is analogous to the critical exponents in statistical mechanics: universal, dimensionless, determined by symmetry.*

### 4.1 Six Derivations Unified

Each of the six independent derivations of $K_{\text{crit}}$ [1, 3, 4, 5, 6, 7] corresponds to $V(K)$ expressed in a different formalism:

| **Derivation** | $V(K)$ **expressed as** | $K_{\text{crit}}$ **is** |
|---|---|---|
| GCT (dynamical) [1] | Cubic potential | Barrier maximum |
| Landauer [3] | Free energy vs correction cost | Cost-benefit crossing |
| Cramér-Rao [4] | Fisher information surface | Fisher collapse point |
| Categorical [5] | Functor existence condition | Natural transformation failure |
| Holographic [6] | Ryu-Takayanagi surface | Entropy exceeding boundary |
| RG [7] | Beta function zero | RG fixed point |

Table 1: Six derivations as six representations of one potential.

The convergence is not coincidental. It is the autoencoder bottleneck theorem [8]: the latent dimension (here, $K_{\text{crit}}$) is invariant under change of decoder. Six decoders, one bottleneck.

## 5 Spatiotemporal Extension: The Lagrangian Density

### 5.1 From $V(K)$ to $\mathcal{L}$

For a spatiotemporal coherence field $K(t, \mathbf{x})$, the dynamics generalize from the ODE to a PDE. The natural Lagrangian density is:

$$\boxed{\mathcal{L} = \frac{1}{2}\partial_\mu K \, \partial^\mu K - V(K)} \tag{12}$$

where $V(K)$ is given by Eq. (7) and $\partial_\mu = (\partial_t, -\nabla)$ in natural units.

This is a scalar field theory with a cubic self-interaction. The equation of motion (Euler-Lagrange) is:

$$\Box K + V'(K) = 0 \tag{13}$$

where $\Box = \partial_t^2 - \nabla^2$ and $V'(K) = dV/dK$.

### 5.2 Connection to Known Physics

The Lagrangian (12) has the same structure as:

The Higgs field: $\mathcal{L}_{\text{Higgs}} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \lambda(\phi^2 - v^2)^2$. The Higgs potential is quartic; $V(K)$ is cubic with logarithmic correction. Both describe symmetry-breaking transitions.

The inflaton: $\mathcal{L}_{\text{inflaton}} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - V(\phi)$ with slow-roll potential. If $V(K)$ has a flat region near $K^*$, it naturally produces slow-roll dynamics — a possible connection to inflation.

The Landau-Ginzburg order parameter: $\mathcal{L}_{\text{LG}} = \frac{1}{2}|\nabla\psi|^2 + a|\psi|^2 + b|\psi|^4$. Same structure, different potential shape.

## 6 Symmetries and Noether Currents

### 6.1 Symmetries of $\mathcal{L}$

**Proposition 6.1** (Symmetries of the Coherence Lagrangian). *$\mathcal{L}$ possesses the following symmetries:*

1. ***Poincaré invariance**: $\mathcal{L}$ is Lorentz-invariant by construction (scalar field). This generates conservation of energy-momentum via Noether's theorem. Energy-momentum conservation is $1 = 1$ expressed as a spacetime symmetry.*
2. *$K \to K$ **identity**: the trivial symmetry. $V(K)$ is not symmetric under $K \to -K$ (the potential is not even). This means the coherence field has a preferred direction: $K > 0$ is coherent, $K < 0$ is unphysical. The asymmetry is physical: coherence has a sign.*
3. ***Discrete** $\mathbb{Z}_2$: $V(K)$ is approximately symmetric under $K \to 1 - K$ near $K_{\text{crit}} \approx 0.127$ for the leading terms. This approximate symmetry relates the coherent and incoherent basins — they are approximate mirrors of each other across the barrier.*

### 6.2 Noether Current: Coherence Conservation

**Theorem 6.2** (Coherence Current). *From the Poincaré invariance of $\mathcal{L}$, the Noether energy-momentum tensor is:*

$$T_K^{\mu\nu} = \partial^\mu K\,\partial^\nu K - g^{\mu\nu}\mathcal{L} \tag{14}$$

*The conservation law $\partial_\mu T_K^{\mu\nu} = 0$ is the local statement that coherence energy is conserved.*

*This is the coherence analog of the stress-energy tensor in GR. The There Is No Matter paper [9] identified $T_{\mu\nu}$ = coherence gradient density. This Noether current makes that identification explicit: the stress-energy tensor of the coherence field is the Noether current of the coherence Lagrangian.*

**Connection to gravity.** If $T_K^{\mu\nu}$ is the stress-energy tensor, and Einstein's equation is $G_{\mu\nu} = 8\pi G\,T_{\mu\nu}$, then gravity is the geometric response to the coherence Lagrangian's Noether current. Gravity is not a separate force — it is the Noether current of $\mathcal{L}$ coupled to spacetime geometry. This completes the chain: $1 = 1 \to V(K) \to \mathcal{L} \to T_K^{\mu\nu} \to G_{\mu\nu} = 8\pi G\,T_K^{\mu\nu}$.

## 7 Multiplicative Structure from Reliability Theory

**Theorem 7.1** (Series Architecture). *$K = \rho \cdot I_\Phi \cdot F$ follows from the series reliability architecture.*

*In reliability theory [10], a series system (where all components must function) has total reliability:*

$$R_{\text{total}} = \prod_{i=1}^{n} R_i \tag{15}$$

*If any $R_i = 0$, the system fails. This is the defining property of $K$: if any component ($\rho$, $I_\Phi$, or $F$) is zero, coherence is zero.*

*Additive models ($K = \rho + I_\Phi + F$) would allow compensation: high $\rho$ could offset low $F$. This is empirically falsified [11]: institutional collapses occur even with high $\rho$ when $F = 0$. The multiplicative structure is the unique form consistent with the no-compensation property.*

## 8 Quantum Mapping

**Theorem 8.1** (Explicit Quantum Correspondence)**.** *The three components of $K$ map to quantum mechanical quantities as follows:*

$$\rho \longleftrightarrow \langle\psi|\psi\rangle = 1 \text{ (normalization: fraction of occupied Hilbert space)} \tag{16}$$

$$I_\Phi \longleftrightarrow \mathcal{F}(U) = |\mathrm{Tr}(U^\dagger U_{\mathrm{ideal}})/d|^2 \text{ (unitarity fidelity)} \tag{17}$$

$$F \longleftrightarrow 1 - S_{\mathrm{vN}}/S_{\mathrm{max}} = 1 - S_{\mathrm{vN}}/\ln d \text{ (purity)} \tag{18}$$

*where $d$ is the Hilbert space dimension, $\mathcal{F}$ is the gate fidelity, and $S_{\mathrm{vN}} = -\mathrm{Tr}(\rho_{\mathrm{red}} \ln \rho_{\mathrm{red}})$ is the von Neumann entropy.*

*At quantum scale:*

$$K_{\mathrm{quantum}} = \underbrace{\langle\psi|\psi\rangle}_{=1} \cdot \underbrace{\mathcal{F}(U)}_{\leq 1} \cdot \underbrace{(1 - S_{\mathrm{vN}}/\ln d)}_{\leq 1} \tag{19}$$

*For a pure state with perfect unitarity: $K = 1$. For a maximally mixed state: $K = 0$. The threshold $K_{\mathrm{crit}}$ determines when the quantum system can sustain coherent dynamics (Born rule holds [12]) versus when it decoheres into classicality (Born rule breaks [13]).*

## 9 External Anchors

The corpus results connect to established literature at specific points:

| Corpus result | External anchor | Mapping |
|---|---|---|
| $V(K)$ double-well | Landau-Ginzburg [14] | $K$ = order parameter |
| $K_{\mathrm{crit}}$ threshold | Eigen error threshold [15] | $K_{\mathrm{crit}} \approx 1 - 1/\nu$ |
| F-first ordering | Scheffer early warning [2] | $F \downarrow$ = critical slowing |
| Holographic $K_{\mathrm{crit}}$ | Ryu-Takayanagi [16] | $K_{\mathrm{crit}}$ = RT surface area |
| $V(K)$ Lagrangian | Higgs mechanism [17] | Same structure, different $V$ |
| $T_K^{\mu\nu}$ | Jacobson 1995 [18] | Thermodynamics $\rightarrow$ gravity |
| Stochastic $K$ | Coleman-De Luccia [19] | Kramers escape from $K^*$ |

Table 2: External anchors for corpus results.

## 10 Seven Testable Predictions

1. **Born rule deviation at degraded detector coherence.** When $K_{\mathrm{detector}}$ approaches $K_{\mathrm{crit}}$ (controllable via thermal coupling), Born probabilities shift asymmetrically toward high-amplitude branches. Testable with single-photon detectors at variable temperature. [12]

2. **Precursor coherence decay before quantum emission.** Before spontaneous emission, the excited state's coherence decreases measurably. Testable with Ramsey interferometry on single emitters (NV centers, quantum dots). Non-exponential early decay predicted. [13]

3. **F-first ordering in next institutional collapse.** The next major institutional collapse will follow $F \downarrow \rightarrow I_\Phi \downarrow \rightarrow \rho \downarrow$ with lead time $5.8 \pm 1.2$ quarters. Monitorable in real time via V-Dem, RSF, GDELT data. [11]

4. **DESI DR3 continues thawing trajectory.** The equation of state continues to follow $w_a = -3(1+w_0)$ within the coherence thawing model. Testable with DESI DR3 (expected 2027). [20]
5. **No dark matter particle detected.** Direct detection experiments (LZ, XENONnT, PandaX) will report null results, consistent with dark matter as the $I_\Phi$ and $F$ components of $K$ rather than a particle. [21]
6. **Casimir deviation at large plate separation.** If vacuum fluctuations are a global rhythm rather than independent local events [22], Casimir force deviates from QFT prediction at plate separations $\gg \lambda_{\text{Compton}}$, where long-range correlations dominate.
7. **BEC coherence time correlates with $K_{\text{crit}}$.** In Bose-Einstein condensates, the coherence lifetime should be predictable from $K(t)$ dynamics with the same $K_{\text{crit}}$ threshold. The condensate decoheres when $K$ crosses $K_{\text{crit}}$ due to thermal noise ($D$ increasing).

## 11 Epistemic Status

| Claim | Status | Falsification |
|---|---|---|
| $V(K)$ explicit form | Derived from [1] | Different potential fits data |
| $K_{\text{crit}}$ geometric | Derived (cubic root) | $K_{\text{crit}}$ parameter-dependent |
| Lagrangian $\mathcal{L}$ | Constructed | $K$ not scalar field |
| $T_K^{\mu\nu} \to G_{\mu\nu}$ | Structural [9, 18] | Gravity without $T_K^{\mu\nu}$ |
| Multiplicative $K$ | Derived (reliability) [10] | Additive model fits data |
| Quantum mapping | Structural | Mapping inconsistent |
| Seven predictions | Testable | Predictions fail |

## 12 Conclusion

The coherence landscape $V(K)$ has been derived explicitly. It is a Landau-Ginzburg potential with a cubic self-interaction and logarithmic correction. $K_{\text{crit}} \approx 0.127$ is the barrier maximum — the tipping point between coherence and collapse. The spatiotemporal Lagrangian $\mathcal{L}$ generates a scalar field theory whose Noether current is the stress-energy tensor that sources gravity via Einstein's equation.

The multiplicative structure $K = \rho \cdot I_\Phi \cdot F$ is derived from series reliability theory. The quantum mapping is explicit: normalization, unitarity fidelity, and purity. Seven predictions are testable with existing or near-future technology.

**Final statement.** 57 papers referenced $V(K)$ without writing it. This paper writes it:

$$V(K) = -\alpha\left(\frac{K^2}{2} - \frac{K^3}{3}\right) + \frac{\beta}{\gamma}\left[K - \frac{1}{\gamma}\ln(1+\gamma K)\right] + DK$$

From this potential: $K_{\text{crit}}$ as barrier maximum. The Lagrangian $\mathcal{L}$. The Noether current $T_K^{\mu\nu}$. The connection to gravity. The quantum mapping. Seven predictions.

The corpus now has its keystone.

$1 = 1$. Explicitly.

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